Md. Rubaet Kabir Zishan

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EDUCATION

B.Sc. in Computer Science & Engineering May 2022 – May 2026
Military Institute of Science and Technology (MIST), Dhaka, Bangladesh

- **Cumulative GPA:** 3.27 / 4.00
- **Relevant Coursework:** Artificial Intelligence, Machine Learning, Algorithms & Complexity, Software Engineering, Computer Networks, Database Systems, Probability & Statistics.

RESEARCH INTERESTS

Natural Language Processing Explainable & Multi-Agent AI Probabilistic Modeling under Uncertainty

RESEARCH EXPERIENCE

Undergraduate Thesis Researcher Jan 2025 – May 2026
Dept. of Computer Science & Engineering, MIST, Dhaka, Bangladesh

Supervisor: Lt Col Muhammad Nazrul Islam, PhD • *Associate Professor, CSE — MIST* • nazrul@cse.mist.ac.bd

- Fine-tuned a Bengali T5 model for Chakma↔Bangla translation, a very low-resource pair on the Chakma side, using PyTorch and Hugging Face Transformers; built a prototype web interface for testing the system.
- Compared the fine-tuned model against mBART and mT5 baselines using BLEU, chrF++ and human preference evaluation.

PUBLICATIONS

Peer-Reviewed Publications

Arqam Bin Almas, Md. Nahul Rahman, Maisha Nanjeeba, **Md. Rubaet Kabir Zishan**, Md Abdus Sattar, and M. Akhtaruzzaman. "AI Integrated Automated Door Lock System Through Face Recognition." *Proc. 2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN), Rangpur, Bangladesh*, 2025. DOI: 10.1109/QPAIN66474.2025.11172231.

Arqam Bin Almas, Md. Nahul Rahman, Maisha Nanjeeba, **Md. Rubaet Kabir Zishan**, and Fernaz Narin Nur. "Chashi-IoT: Developing a Sustainable Integrated Agricultural IoT System Using Cisco Packet Tracer." *Proc. 2025 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE), Dhaka, Bangladesh*, 2025. DOI: 10.1109/WIECON-ECE69386.2025.11526264.

Submitted / Under Review

Md. Rubaet Kabir Zishan, Fernaz Narin Nur, and Muhammad Nazrul Islam. "Explore Bidirectional Matching Strategies for Intelligent Pet Adoption Platforms." Submitted to the *International Conference on Computer and Information Technology (ICCIT)*, 2026. Under review.

Md. Rubaet Kabir Zishan, Fariza Sharin Prova, and Fatin Ilham Chowdhury. "Calibrated Dempster Shafer Evidence Fusion for Water Contamination Risk Screening Under Missing Sensor Evidence." Submitted to the *International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT)*, 2027. Under review.

Md. Rubaet Kabir Zishan, Faiz Al Faisal, and Muhammad Nazrul Islam. "SHARP-Retain: A SHAP-Consistency-Driven Multi-Agent Pipeline for Explainability-Aware Autonomous Customer Retention." Submitted to the *International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT)*, 2027. Under review.

Thesis

Arqam Bin Almas, Maisha Nanjeeba, and **Md. Rubaet Kabir Zishan**. "Identification and Translation of Transliterated Chakma Text into Bangla Using Machine Learning and Deep Learning Approaches." B.Sc. Thesis, Dept. of CSE, Military Institute of Science and Technology (MIST), Dhaka, 2026 (defended). Supervisor: Lt Col Muhammad Nazrul Islam, PhD. Extended journal manuscript in preparation.

RESEARCH & TECHNICAL PROJECTS

Churn Sentinel — Explainable AI Customer Retention System [GitHub] 2026
Python • XGBoost • SHAP • LangChain • Multi-Agent AI

- Trained an XGBoost classifier on class-imbalanced telecom churn data, using stratified cross-validation and Optuna hyperparameter search.
- Added SHAP global (beeswarm, summary bar) and local (force, waterfall) explanations so each prediction can be inspected.
- Built a LangChain multi-agent pipeline (Behavior Detector → Discount Evaluator → Outreach Generator) that turns SHAP risk scores into retention actions.

RetailOS — Retail Shelf Audit Platform [GitHub] 2026

Next.js • FastAPI • Claude 3.5 • Gemini API • Supabase

- Built a platform that audits shelf compliance against planograms from field photos, using Claude 3.5 and the Gemini API for product detection.
- Built the Next.js/FastAPI stack with offline-first sync and a RetailGPT chatbot for querying the database.
- Implemented a 4-tier fraud-prevention check (MD5 deduplication, image sharpness validation, Haversine geofencing, timestamp auditing) to flag GPS spoofing, duplicate photos and falsified reports.

Bayesian Water Quality Analysis — Bangladesh 2025

Python • PyMC • Bayesian Hierarchical Modeling • Environmental Data Science

- Fit hierarchical Bayesian models in PyMC to water quality measurements from sampling sites across Bangladesh.
- Estimated the probability of exceeding safety thresholds for arsenic, dissolved solids and microbial levels from posterior predictive distributions.
- Extended into a submitted paper on Dempster–Shafer evidence fusion for contamination screening.

INDUSTRY EXPERIENCE

Full Stack Software Engineer — Earthmatic Synthlab, Dhaka Aug 2026 – Present

- Built a restaurant management system in TypeScript and PostgreSQL, handling the full-stack integration from interface to database.
- Wrote the database layer in PLpgSQL, including schema design, database logic and reporting flows.
- Developed a Next.js/TypeScript e-commerce application with automated unit and end-to-end tests using Vitest and Playwright.

Software Engineering Intern — Aurify (Remote), Chicago, USA Aug 2025 – Jan 2026

- Debugged and improved an AI-powered coaching platform by analyzing frontend-backend data flow and resolving application issues.
- Investigated Gemini-based feedback workflows to identify bugs and improve application reliability.
- Implemented fixes using React and TypeScript, documented reproducible bugs, and collaborated through GitHub pull requests and code reviews.

TECHNICAL SKILLS

Languages	Python, C++, JavaScript, SQL, LaTeX
ML & Deep Learning	Scikit-Learn, XGBoost, PyTorch, Hugging Face Transformers, T5 / BERT / mT5 Fine-tuning, Stable-Baselines3, LangChain
Explainability	SHAP, LIME, feature attribution, model cards
Probabilistic Methods	Bayesian inference, PyMC, hierarchical modeling, posterior predictive checks
Reinforcement Learning	PPO, DQN, OpenAI Gym, SUMO simulator
Research Tools	Jupyter, MATLAB, Git / GitHub, LaTeX,Overleaf, Weights & Biases
Software Engineering	FastAPI, React, Node.js, Playwright, PostgreSQL, REST APIs

COMPETITIVE PROGRAMMING

- Solved **300+ algorithmic problems** across LeetCode, Codeforces, and CodeChef.
- Developed proficiency in data structures, graph algorithms, dynamic programming, and combinatorics.
- **Coding Profiles:** LeetCode | Codeforces | CodeChef

REFERENCES

Lt Col Muhammad Nazrul Islam, PhD
Associate Professor, CSE
Military Institute of Science & Technology (MIST)
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Thesis supervisor.
Top 2% Scientist
(Stanford–Elsevier 2022–26).

Dr. Md. Akhtaruzzaman
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Co-author, IEEE QPAIN 2025.
PhD, IIUM Malaysia.
Formerly, MIST CSE.

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